

Applied Engineering Optimisation

ENEL445

Power Systems Optimisation

- Term 2 of ENEL445: applications, nature-inspired/evolutionary optimisation.
- This week, I'll give a brief (and not too difficult) overview of optimization in power systems. We'll look at what is used in industry + a brief discussion of future trends.
 - Even if you aren't interested in power systems, treat this as an illustrative example.

Q: How is optimization used in power systems?

- Economic dispatch / Optimal power flow (OPF)
- Power system state estimation
- Also: load forecasting, optimal / model-predictive control of power electronics, scheduling / energy management / demand response, etc.
 - The latter (energy management) is a highly important aspect and links to both forecasting and the optimal power flow problem.

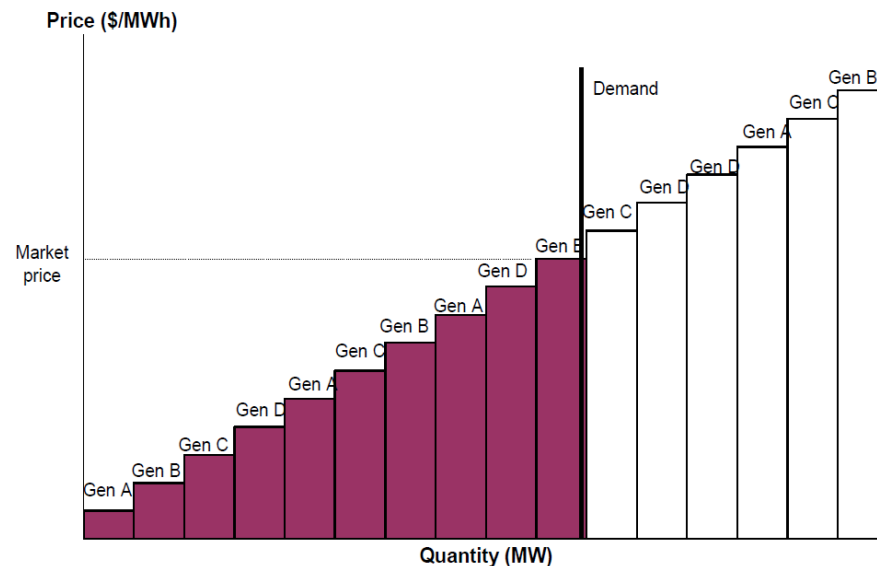
Economic dispatch

- Used in many power markets, e.g. European Power Exchange (EPEX) etc, NZ Electricity Market (until Nov 2022).
- Supply must meet demand, as cheaply as possible, subject to generator limits.
- Does **not** consider any network flows, losses, or network constraints. (Although, *penalty factors* - more on this later - may be used to approximate the effect of losses).

Formulation

Solving the economic dispatch problem

- In the linear case, “merit order” techniques can easily solve the economic dispatch problem, e.g.



- In the non-linear case? Use constrained optimisation techniques!

Economic Dispatch Example

What is economic dispatch for a two generator system $P_D = P_{G1} + P_{G2} = 500$ MW and

$$C_1(P_{G1}) = 1000 + 20P_{G1} + 0.01P_{G1}^2 \quad \$/hr$$

$$C_2(P_{G2}) = 400 + 15P_{G2} + 0.03P_{G2}^2 \quad \$/hr$$

Economic Dispatch Example, cont'd

Inclusion of Transmission Losses

- The losses on the transmission system are a function of the generation dispatch. In general, using generators closer to the load results in lower losses
- This impact on losses should be included when doing the economic dispatch
- Losses can be included by slightly rewriting the Lagrangian:

$$L(\mathbf{P}_G, \lambda) = \sum_{i=1}^m C_i(P_{Gi}) + \lambda(P_D + P_L(P_G) - \sum_{i=1}^m P_{Gi})$$

Impact of Transmission Losses

This small change then impacts the necessary conditions for an optimal economic dispatch

$$L(\mathbf{P}_G, \lambda) = \sum_{i=1}^m C_i(P_{Gi}) + \lambda(P_D + P_L(P_G) - \sum_{i=1}^m P_{Gi})$$

The necessary conditions for a minimum are now

$$\frac{\partial L(\mathbf{P}_G, \lambda)}{\partial P_{Gi}} = \frac{dC_i(P_{Gi})}{dP_{Gi}} - \lambda \left(1 - \frac{\partial P_L(P_G)}{\partial P_{Gi}}\right) = 0$$

$$P_D + P_L(P_G) - \sum_{i=1}^m P_{Gi} = 0$$

Impact of Transmission Losses

Solving each equation for λ we get

$$\frac{dC_i(P_{Gi})}{dP_{Gi}} - \lambda \left(1 - \frac{\partial P_L(P_G)}{\partial P_{Gi}}\right) = 0$$

$$\lambda = \frac{1}{\left(1 - \frac{\partial P_L(P_G)}{\partial P_{Gi}}\right)} \frac{dC_i(P_{Gi})}{dP_{Gi}}$$

Define the penalty factor L_i for the i^{th} generator

$$L_i = \frac{1}{\left(1 - \frac{\partial P_L(P_G)}{\partial P_{Gi}}\right)}$$

Impact of Transmission Losses

The condition for optimal dispatch with losses is then

$$L_1 IC_1(P_{G1}) = L_2 IC_2(P_{G2}) = L_m IC_m(P_{Gm}) = \lambda$$

Since $L_i = \frac{1}{\left(1 - \frac{\partial P_L(P_G)}{\partial P_{Gi}}\right)}$ if increasing P_{Gi} increases

the losses then $\frac{\partial P_L(P_G)}{\partial P_{Gi}} > 0 \rightarrow L_i > 1.0$

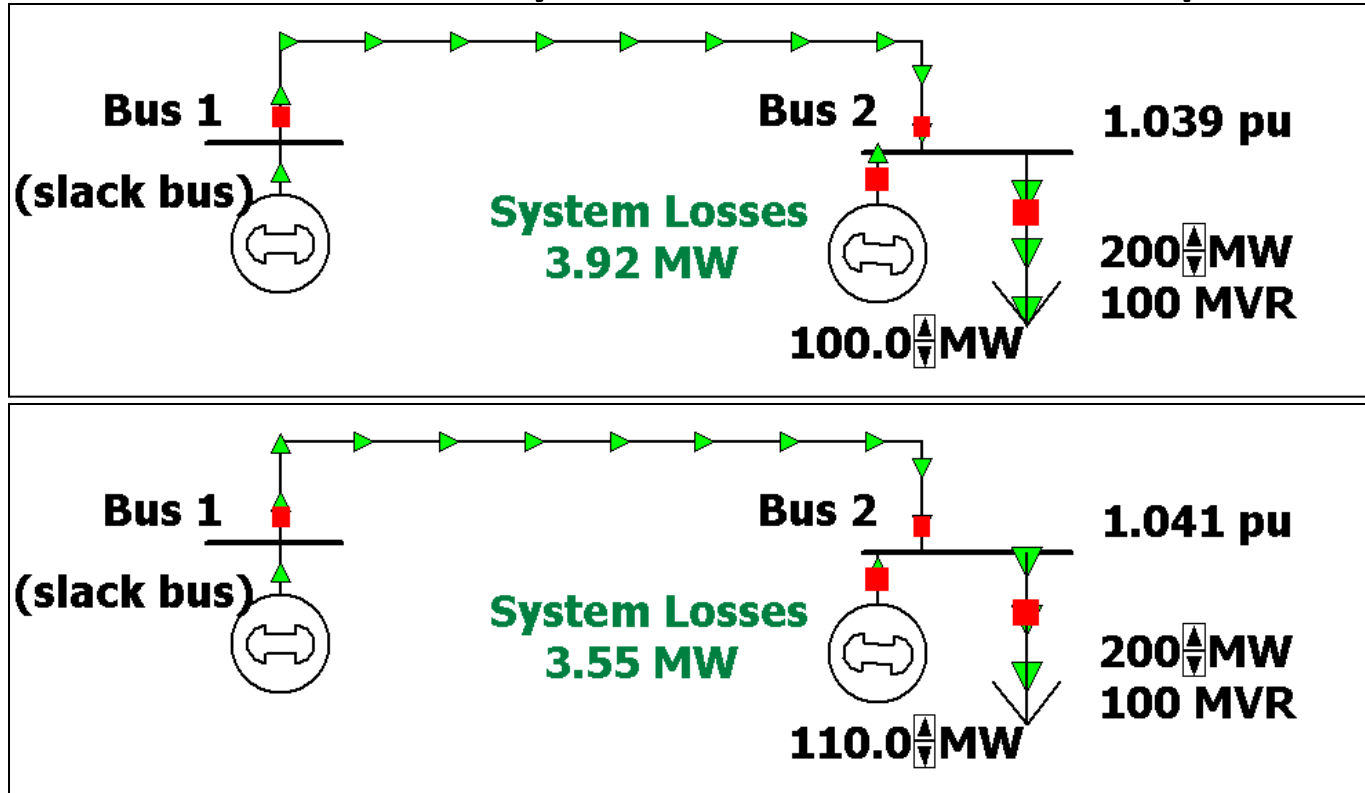
This makes generator i appear to be more expensive (i.e., it is penalized). Likewise $L_i < 1.0$ makes a generator appear less expensive.

Calculation of Penalty Factors

Unfortunately, the analytic calculation of L_i is somewhat involved. The problem is a small change in the generation at P_{Gi} impacts the flows and hence the losses throughout the entire system. However, using a power flow you can approximate this function by making a small change to P_{Gi} and then seeing how the losses change:

$$\frac{\partial P_L(P_G)}{\partial P_{Gi}} \approx \frac{\Delta P_L(P_G)}{\Delta P_{Gi}} \quad L_i \approx \frac{1}{1 - \frac{\Delta P_L(P_G)}{\Delta P_{Gi}}}$$

Two Bus Penalty Factor Example



Variants

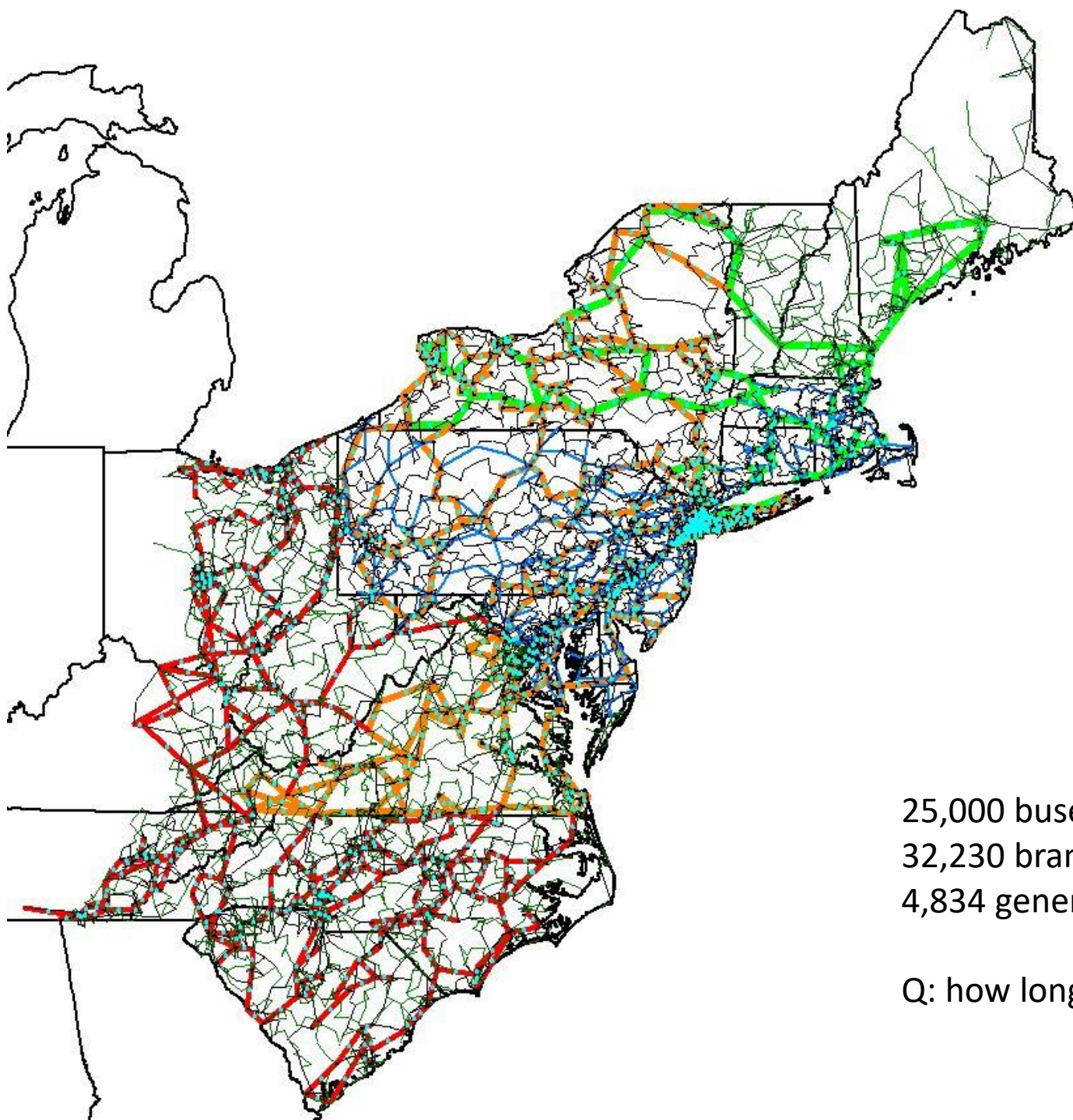
- The two main alternatives are: uniform pricing (e.g. NZ) or pay-as-bid (UK).
 - Uniform pricing (sometimes called *pay-as-cleared*): every generator whose bid is accepted is paid the marginal price (market price in previous slide).
 - Pay-as-bid: every generator whose bid is accepted is paid the price of their bid.
- Can you think of advantages / disadvantages of either?

Optimal power flow

- Given a power system, i.e. a network of generators, power lines, transformers, buses (nodes), and loads ...
- Objective: meet demands at minimum cost
- Cost: generation costs
- Constraints: Power flows follow the laws of physics, limits on equipment etc.

Optimal power flow

- If there are no system constraints, the solution is the same as Economic Dispatch.
- Recall (if you did ENEL382) the power flow equations. Let's formulate the optimal power flow problem:



25,000 buses
32,230 branches
4,834 generators

Q: how long to solve?

Optimal power-flow

- Due to the complexity of the non-linear problem we have posed, most commercial packages and power system operators use a simplified version called (mis-named) DC optimal power flow.
- In the cases where the full AC OPF is solved, by far the best algorithm in practice are *interior-point* solvers.
- Recent studies e.g. by the FERC in the US suggest that a 5-10% reduction of costs could be saved if it were computationally practical to run a full AC OPF.

DC power flow

DC optimal power flow

$$\min \sum_i c_i P_{G_i}$$

subject to:

$$P_{G_i}^{min} \leq P_{G_i} \leq P_{G_i}^{max}$$

$$\mathbf{B} \cdot \boldsymbol{\theta} = \mathbf{P}_G - \mathbf{P}_D$$

$$\frac{1}{x_{ij}}(\theta_i - \theta_j) \leq P_{ij,max}$$

- As an example, let's look at Transpower's DC OPF (solved every 5 minutes for real-time dispatch) [here](#)

DC optimal power flow

- Possible improvements:
 - If we have a better estimate of the voltages, we can use them (constants) instead of 1 p.u.
 - Estimate of losses – typical method:
 - Power flows are calculated via the DC power flow equations
 - An approx. current can be estimated from this (assume 1 p.u. voltage)
 - The resistance of the line – ignored so far – is used to calculate I^2R losses.
 - However, the squaring of the current would create a nonlinear equality constraint - not a linear problem any more – so a piecewise linear approximation is typically used.

DC optimal power flow

- After a solution from the DC OPF is found, a standard AC power flow (note, not OPF) is usually run to ensure the solution is feasible.

OPF types

General problem type	Problem name	Includes voltage angle constraints?	Includes bus voltage magnitude constraints?	Includes transmission constraints?	Includes losses?	Assumptions	Includes generator costs?	Includes contingency constraints?
OPF	ACOPF, or Full ACOPF	Yes	Yes	Yes	Yes		Yes	No
OPF	DCOPF	No	No; all voltage magnitudes fixed	Yes	Maybe	Voltage magnitudes are constant	Yes	No
OPF	Decoupled OPF	Yes	Yes	Yes	Yes	Power-voltage angle are independent of voltage magnitude-reactive power	Yes	No
OPF	Security-Constrained Economic Dispatch (SCED)	Yes	No	Yes	Yes	Voltage magnitudes are constant	Yes	Yes
Power flow	Power Flow, or Load Flow	No, but can be added	Yes	No, but can be added	Yes		No	No
Economic dispatch	Economic Dispatch	No	No	No	Depends	No transmission constraints	Yes	No
OPF	Security Constrained OPF (SCOPF)	Yes	Depends	Yes	Yes	Depends	Yes	Yes

Security-constrained OPF

- Idea: power system dispatch not only has to allow a feasible solution, but also has to satisfy contingency constraints – e.g. if something plausible-but-unexpected happens, the system has to stay within operational limits.
- This is orders of magnitude more expensive than even AC OPF, as the full set of constraints for every contingency needs to be added!

Security-constrained OPF

- What happens in practice is that the DC OPF is solved, and the solution checked against a subset of contingencies (note: this is not solving another OPF but simply plugging the DC OPF solution into a simplified SCOPF and checking for feasibility).
- If the solution is not SCOPF-feasible, heuristic constraints are used to add security and the DC OPF is re-solved, etc. until a SCOPF feasible solution is found.

Locational marginal pricing

- The OPF can be (and is) used to calculate nodal prices (locational marginal prices), for example in NZ.
- As before, the Lagrange multipliers represent prices and these are affected by losses, security constraints, etc.

Summary

- One of the main uses of optimisation in power systems is in optimising dispatch (i.e. the power of each generator).
- Due to computational complexity of the AC OPF / SCOPF, the DC OPF is almost always used in industry, but this may change in the next few decades.

Power system state estimation

- Main idea – work out a best estimate of the states (voltages, currents, power flows) of the power system based on available measurements.
- Could do power-flow but in reality:
 - Might not have 2 measurements per bus (remember ENEL382)
 - Any error is likely to result in non-convergence or useless results
 - Power flow cannot integrate measurements such as P & Q line flows.
- Widely used in the power industry – it is important to know what's happening in the system.
- Often linked to the SCADA / EMSE system.

Formulating state estimation as an optimisation problem

- For simplicity we'll use the DC power flow equations we derived earlier this week.
- Unlike the Optimal Power Flow case, commercial products do run the full AC power-flow equations instead of the “DC” power flow approximation.

Example (from Wood / Wallenberg, chapter 9)

PER UNIT REACTANCES
(100 MVA BASE):

$$X_{12} = 0.2$$

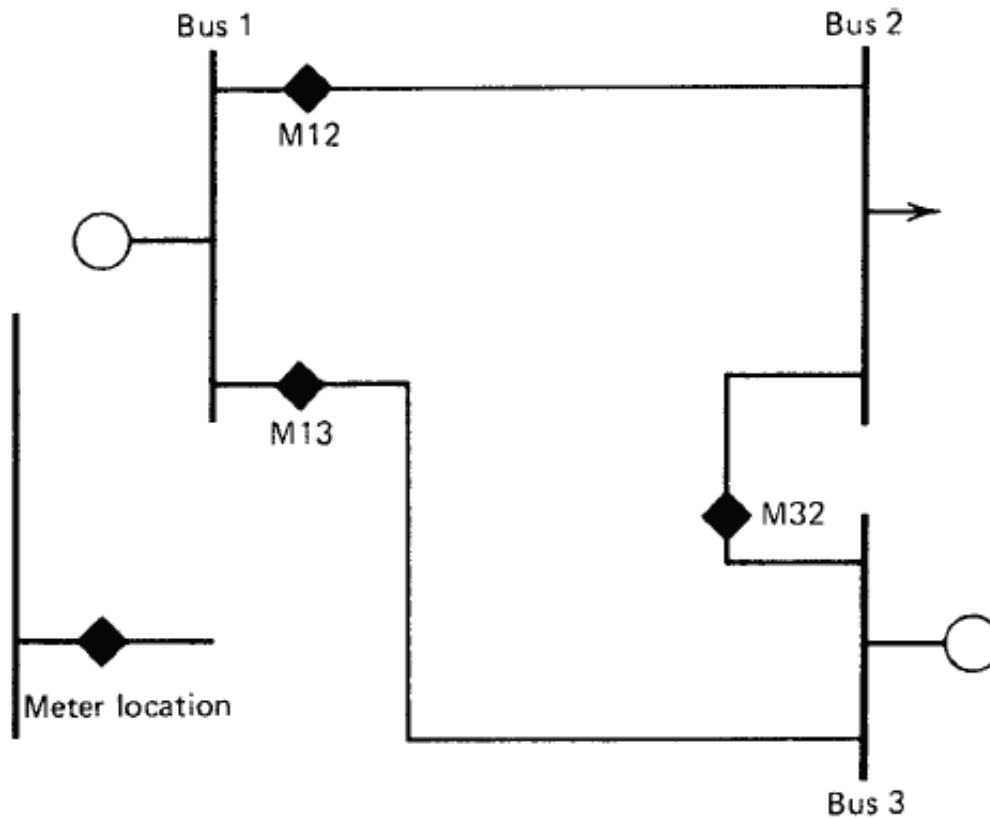
$$X_{13} = 0.4$$

$$X_{23} = 0.25$$

$$M_{12} = 62 \text{ MW} = 0.62 \text{ pu}$$

$$M_{13} = 6 \text{ MW} = 0.06 \text{ pu}$$

$$M_{32} = 37 \text{ MW} = 0.37 \text{ pu}$$



Power system state estimation

- Different meters have different accuracy, so a basic idea is a *weighted* least-squares formulation. We'll go through this (which is the basis for most state estimators in industry).

Formulating state estimation as a weighted least-squares problem

Formulating state estimation as a weighted least-squares problem

Formulating state estimation as a weighted least-squares problem

There is also a closed-form solution to the problem when $N_s > N_m$, although in this case we are not estimating $\mathbf{x}$ to maximize a likelihood function since $N_s > N_m$ usually implies that many different values for $\mathbf{x}^{\text{est}}$ can be found that cause $f_i(\mathbf{x}^{\text{est}})$ to equal z_i^{meas} for all $i = 1, \dots, N_m$ exactly. Rather, the objective is to find $\mathbf{x}^{\text{est}}$ such that the sum of the squares of x_i^{est} is minimized. That is,

$$\min_{\mathbf{x}} \sum_{i=1}^{N_s} x_i^2 = \mathbf{x}^T \mathbf{x} \quad (9.25)$$

subject to the condition that $\mathbf{z}^{\text{meas}} = [\mathbf{H}]\mathbf{x}$. The closed-form solution for this case is

$$\mathbf{x}^{\text{est}} = [\mathbf{H}]^T \left[[\mathbf{H}][\mathbf{H}]^T \right]^{-1} \mathbf{z}^{\text{meas}} \quad (9.26)$$

Formulating state estimation as a weighted least-squares problem

TABLE 9.1 Estimation Formulas

Case	Description	Solution	Comment
$N_s < N_m$		$\mathbf{x}^{\text{est}} = [[H]^T[R^{-1}][H]]^{-1} \times \{H\}^T[R^{-1}]\mathbf{z}^{\text{meas}}$	$\mathbf{x}^{\text{est}}$ is the maximum likelihood estimate of $\mathbf{x}$ given the measurements $\mathbf{z}^{\text{meas}}$
$N_s = N_m$	Completely determined	$\mathbf{x}^{\text{est}} = [H]^{-1}\mathbf{z}^{\text{meas}}$	$\mathbf{x}^{\text{est}}$ fits the measured quantities to the measurements $\mathbf{z}^{\text{meas}}$ exactly
$N_s > N_m$	Underdetermined	$\mathbf{x}^{\text{est}} = [H]^T[[H][H]^T]^{-1}\mathbf{z}^{\text{meas}}$	$\mathbf{x}^{\text{est}}$ is the vector of minimum norm that fits the measured quantities to the measurements exactly. (The norm of a vector is equal to the sum of the squares of its components)

Power system state estimation

- In power system state estimation, underdetermined problems are not solved.
- Instead, “pseudo-measurements” are added to the measurement set to give a completely determined or over-determined problem.
- These pseudo-measurements in the past were obtained by telephone call to the generator operators! They can also be based on historical / previous measurements. A large standard deviation is assigned as we have little confidence in the “pseudo-measurement”.

Example (from Wood / Wallenberg, chapter 9)

PER UNIT REACTANCES
(100 MVA BASE):

$$X_{12} = 0.2$$

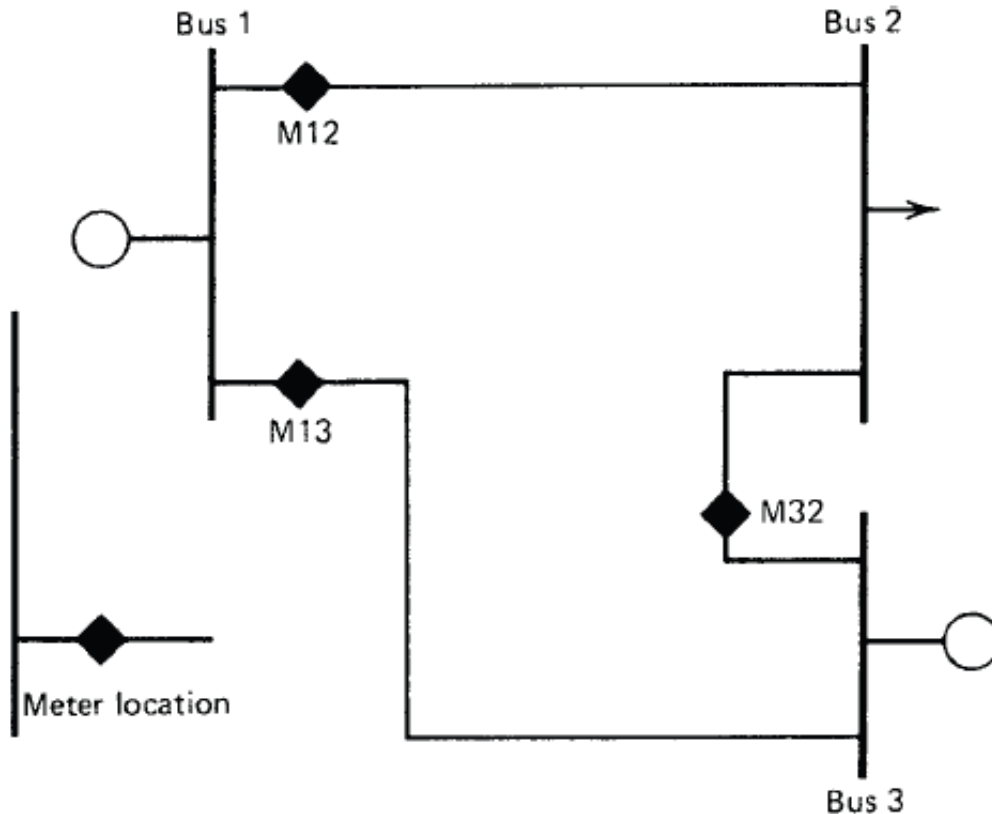
$$X_{13} = 0.4$$

$$X_{23} = 0.25$$

$$M_{12} = 62 \text{ MW} = 0.62 \text{ pu}$$

$$M_{13} = 6 \text{ MW} = 0.06 \text{ pu}$$

$$M_{32} = 37 \text{ MW} = 0.37 \text{ pu}$$



To start, we will assume that all three meters have the following characteristics.

Meter full-scale value: 100 MW

Meter accuracy: ± 3 MW

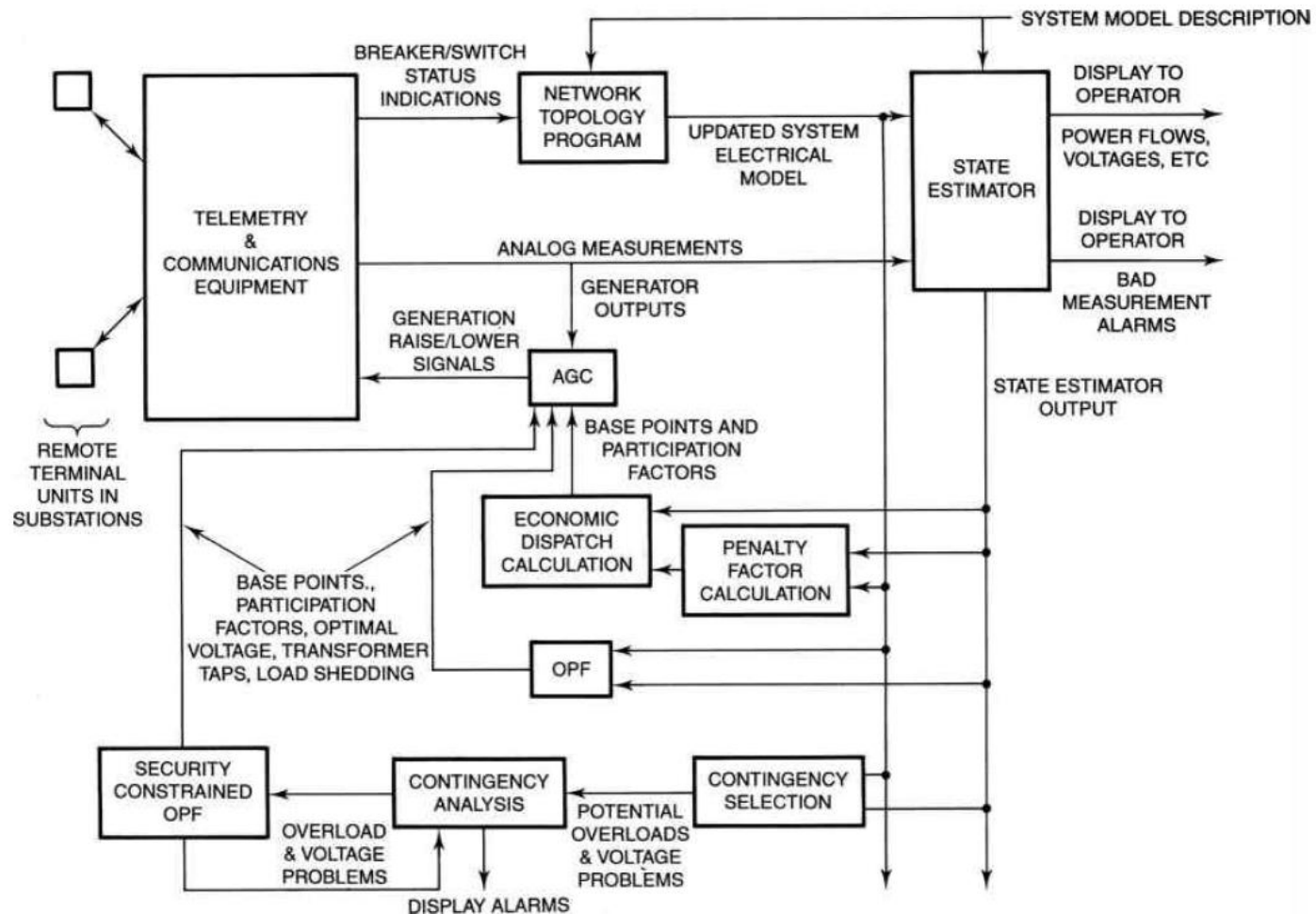
Example

Example

Practical considerations

- Generally, the nonlinear AC power-flow equations are solved. A common method is to use Newton's method.
- Phasor Measurement Units using GPS synchronization are becoming increasingly common and improving state estimation accuracy – but the state estimation problem is also getting harder due to inverter-based resources.
- More advanced topics:
 - New algorithms to cope without enough measurements (in general: more generators = more measurements required), or without exact network data.
 - Detecting and removing bad data.

How it all fits together



Power system state estimation

- Further resources:
 - Wood / Wallenberg textbook
 - More detailed lecture notes:
<http://www.ee.unlv.edu/~eebag/State%20Estimation.pdf>
 - https://www.kth.se/social/upload/518a08d3f27654786295ca51/Lecture_15_StateEstimation.pdf
 - <https://ieeexplore.ieee.org/document/4596092> (New England example)
 - <https://home.engineering.iastate.edu/~jdm/ee553/SE1.pdf>
 - Lots of journal and conference papers on IEEE, etc.